

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> First let's calculate how long it takes to download 40% of the file . Since the file size is 200 GB . 40% of 200 GB is $0.4 * 200 = 80$ GB . Given that she can download at a speed of 2 GB / minute . To download 80 GB would take $80 / 2 = 40$ minutes . Now let's consider the total time spent before the restart . Initially downloading 80 GB takes 40 minutes . Then the restart process takes an additional 20 minutes . In total $40 + 20 = 60$ minutes . After the restart the file download has to start from the beginning again . The file size is 200 GB . At a speed of 2 GB / minute . Downloading 200 GB would take $200 / 2 = 100$ minutes . Therefore in total Carla spends $60 + 100 = 160$ minutes to complete the download . </think><answer> 160 </answer>

